ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CE ENGINEERING — SEMESTER 4

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Differentiate Static vs Kinematic Indeterminacy. State Castigliano's First and Second Theorems. [14 Marks]

[OR]

Q.1 (Alt) State Unit Load Method and Maxwell's Reciprocal Theorem with proofs. [14 Marks]

--- UNIT II ---

Q.2 Analyze propped cantilever beam using strain energy method. State Clapeyron's Three Moment Theorem. [14 Marks]

[OR]

Q.2 (Alt) Analyze two-span continuous beam using Theorem of Three Moments. Draw SFD and BMD. [14 Marks]

--- UNIT III ---

Q.3 Derive Slope Deflection Equations for members with and without support settlement. [14 Marks]

[OR]

Q.3 (Alt) Analyze single-bay portal frame using Slope Deflection Method. [14 Marks]

--- UNIT IV ---

Q.4 Explain Moment Distribution Method (Hardy Cross). Carry-over, Stiffness, Distribution factors. [14 Marks]

[OR]

Q.4 (Alt) Analyze symmetrical portal frame with sway using Moment Distribution Method. [14 Marks]

--- UNIT V ---

Q.5 Draw Influence Line Diagrams (ILD) for reactions and bending moments in simply supported beams. [14 Marks]

[OR]

Q.5 (Alt) State Muller-Breslau Principle. Analyze Three-Hinged Parabolic Arch horizontal thrust. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****